Homogenous and Isotropic Turbulence

UE 9TE02 - Turbulence - CFD avancée

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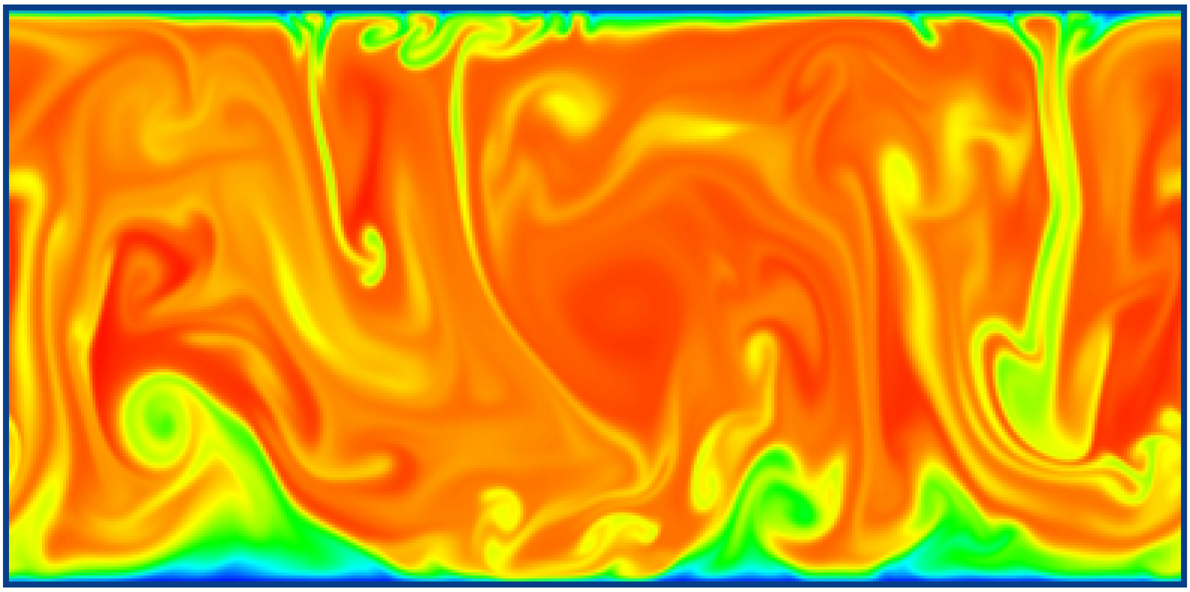

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Resume

Nous avons mené des expériences avec la soufflerie à circuit ouvert du laboratoire PRISME, qui possède une section d'essai de 2 m de long et une section transversale de 0,5 m × 0,5 m. Le but de ces expériences est d'étudier la turbulence homogène et isotrope, ainsi que de créer une base de données expérimentale en fonction de diverses grilles, avec des solidités différentes, qui seront installées dans la veine d'essai.

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1 Introduction

The study of external flows along flat plates is crucial in numerous applications within the aeronautical and aerospace domains. Indeed, the boundary layer develops comparably on a flat plate as it does on 2D and 3D shaped bodies (for example : aircraft wings, fuselage, etc.). The boundary layer plays a major role in the performance of a lifting surface. For instance, it dictates the aerodynamic performance of an aircraft, determining both the lift and drag of a wing, which can be considered as a flat plate.

A boundary layer is the interface region between a solid surface and the surrounding fluid during relative motion between them. At this interface, a zero relative velocity is observed, known as the condition of adhesion, resulting from the fluid's viscosity. The very definition of the boundary layer lies in the fact that it represents the flow region where viscous effects are at least as significant as inertial effects (in terms of order of magnitude). This is not the case far from the wall, where viscous effects are practically negligible. The boundary layer consists of 3 regimes :

Laminar Transitional Turbulent The boundary layer will be studied using a hot-wire anemometer, which involves maintaining a constant temperature (higher than the ambient temperature) of a very fine metal wire. After calibration, this device allows for precise measurement of velocity fluctuations at a specific point. The calibration of the hot-wire will be performed using a Pitot tube placed under the flat plate. To study the evolution and properties of the boundary layer, the hot-wire is mounted on a manual displacement system allowing movements in the direction of the main flow (x) and perpendicular to the wall (y). A digital calliper enables precise control of the probe's positioning along y.

The aim is to identify the laminar-to-turbulent transition and compare the properties of the boundary layer in these two regimes.

This report consists of three parts. The principle of the boundary layer as well as the formulated assumptions will be explained in the first section. The experimental setup used during the study will be detailed in the second part. Finally, the results obtained from the experimental trials will be analysed and discussed in the last section.

2 Measurement campaign

The assumptions made for this lab experiment are as follows : - The boundary layer effects along the walls of the wind tunnel itself will be neglected during the study.

First, we will calibrate our experimental setup using King's Law :

$$E^2 = A \cdot U^{0.45} + B$$

Where : - A and B are constants that need to be determined. - E is the voltage. - U is the velocity calculated from the pressure differential measured by the Pitot probes located in the wind tunnel, given by :

$$U = \sqrt{2\frac{gh}{\rho_{\text{air}}}}$$

A boundary layer is a layer of fluid that forms around an object when it is placed in a flow. This effect is caused by the fluid's viscosity and results in a decrease in fluid velocity at the contact point with the object.

For a laminar boundary layer, there are two characteristic times that are approximately equal :

- Advection time : $\frac{x}{U_\infty}$ - Viscous time : $\frac{\delta^2}{\nu}$

Since these values are approximately equal, we can write :

$$\frac{\frac{x}{U_\infty}}{\frac{\delta^2}{\nu}} \approx 1$$

$$\delta^2 \approx A\frac{x \cdot \nu}{U_\infty}$$

$$\delta \approx A\sqrt{\frac{x^2 \cdot \nu}{x \cdot U_\infty}}$$

$$\delta \approx A\sqrt{x} \cdot Re^{-\frac{1}{2}}$$

The thickness of the boundary layer follows this equation, so the logarithm of x should have a slope of 0.5.

A turbulent boundary layer forms after a critical value of x on an object, creating turbulence within this layer.

2.1 Grid Selection and Mesh Estimation

The purpose of this experiment is to study isotropic turbulence. To achieve this, we needed to install a grid in our wind tunnel to disturb the flow by creating shear in the flow. Each grid, characterized by its mesh size M and solidity σ (the ratio between the blocked area and the total area), can be installed at the inlet of the test section.

The velocity field will be measured using a two-component hot-wire probe, which is mounted on a traverse allowing 2D motions in the (x, y) plane. This setup allows simultaneous probing of both the streamwise and spanwise velocity components, $u(x, y)$ and $v(x, y)$, respectively. Additionally, a thermocouple is used to monitor temperature variations to compensate for thermal drift.

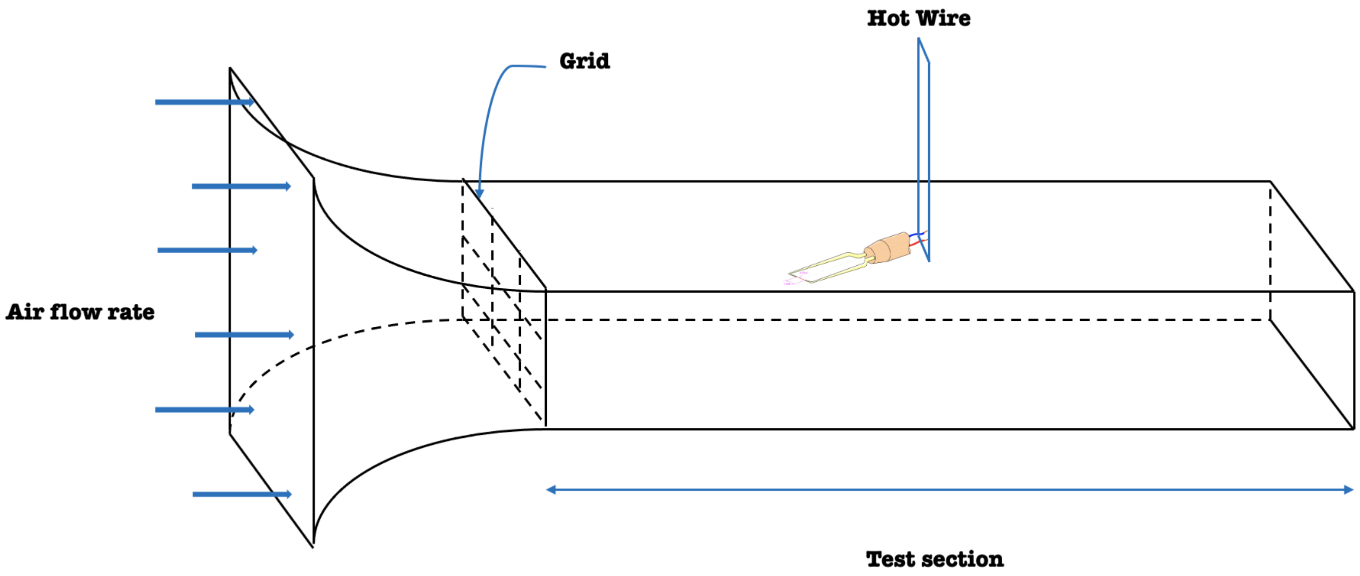

Figure 1 – Loi de King low

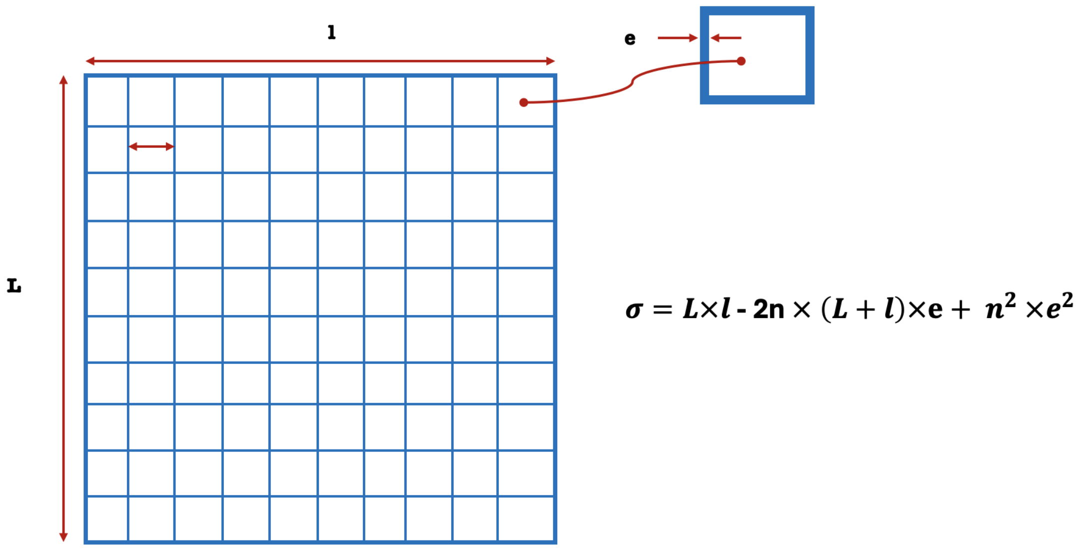

Figure 2 – Grid

2.2 Probe Calibration

To record the various flow velocities, we will use a computer and a data acquisition software managed by the labVIEW software called MesureFilChaud.

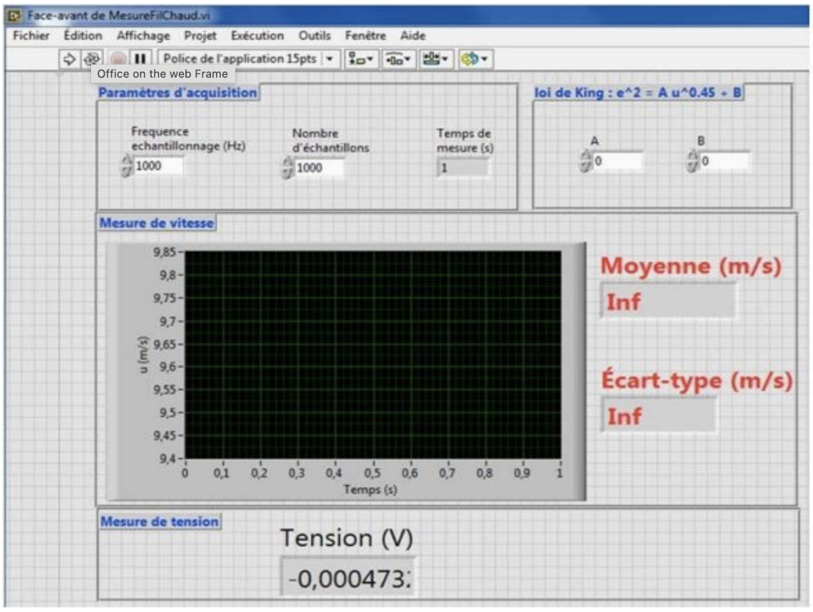

FIGURE 3 – Software MesureFilChaud

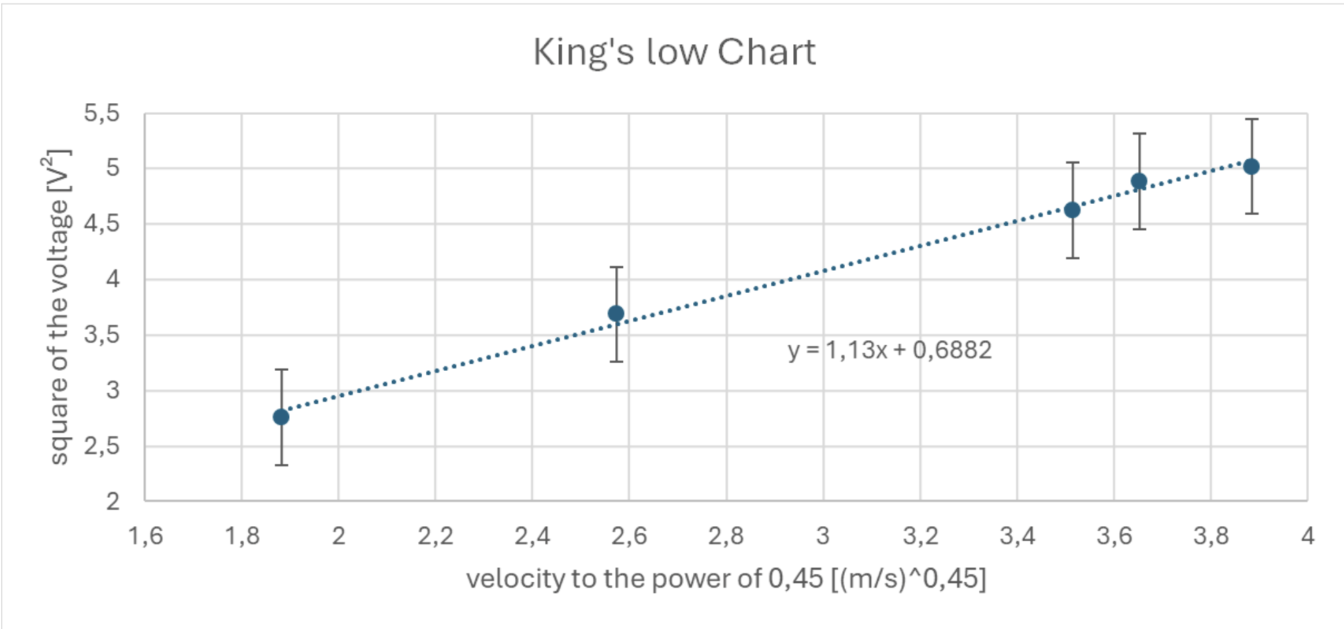

FIGURE 4 – Loi de King low

we will start by calibrating the hot-wire using the previously mentioned King's Law, aiming to find the appropriate coefficients A and B. To determine these coefficients, we varied the flow velocity within the wind tunnel. For each value, we recorded the corresponding voltage. Then, we plotted a graph of voltage squared against flow velocity. This allowed us to establish a trendline and derive average values for coefficients A and B, minimizing errors.

3 One-Point Statistics

3.1 Mean Velocity Components

Compute the mean velocity components $U(x,y)$ and $V(x,y)$. How do these quantities evolve along the centerline of the wind tunnel ?

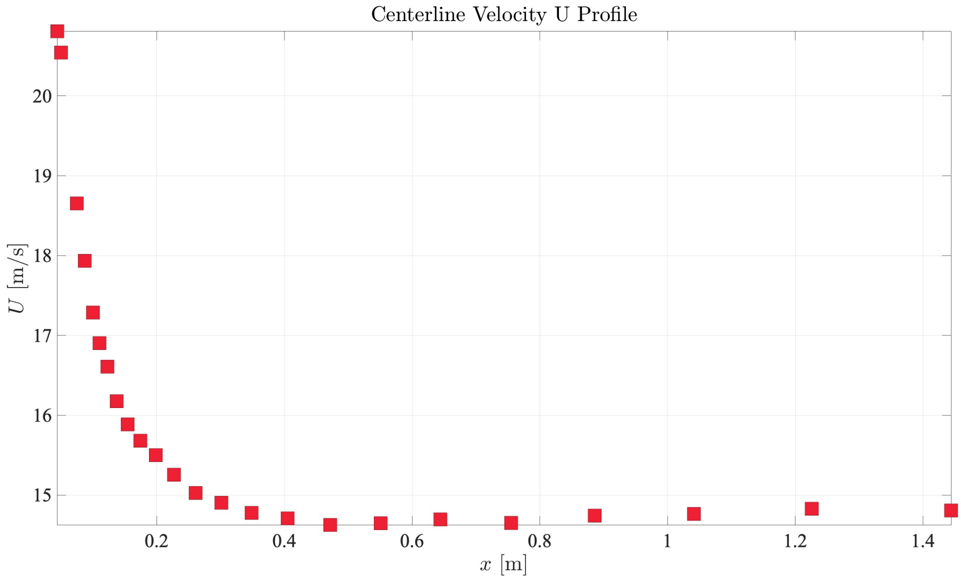

FIGURE 5 – Mean velocity U

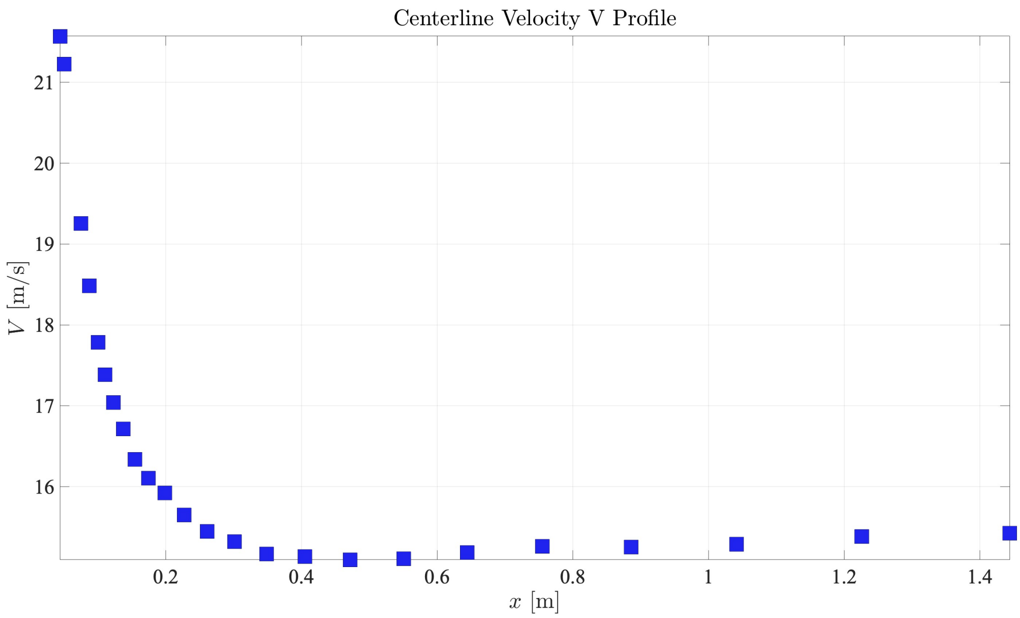

FIGURE 6 – Mean velocity V

The velocity profiles $U(x,y)$ and $V(x,y)$, shown in Figures and respectively, represent the evolution of the velocity components along the centerline of the flow for two orthogonal directions.

— **Figure : Centerline Velocity U Profile**
The streamwise velocity component, $U(x,y)$, decreases rapidly near the inlet and continues to diminish as x increases. This sharp decay indicates that the turbulent structures formed by the

grid dissipate quickly, leading to a rapid loss of momentum along the centerline. The flow tends to stabilize at lower velocity values downstream, which implies that the turbulence intensity weakens as the flow progresses.

— **Figure : Centerline Velocity V Profile**
The spanwise velocity component, $V(x,y)$, exhibits a similar trend to the streamwise velocity, with a steep drop-off near the inlet. This behavior suggests that turbulence is more isotropic in the near field but becomes increasingly organized in the downstream region. The spanwise velocity becomes almost negligible further downstream, signifying that the primary contribution to the overall flow is from the streamwise direction as the turbulence decays.

In both cases, the velocity profiles exhibit a clear dependence on the distance from the inlet, with stronger velocity fluctuations near the grid, followed by a decay as the flow approaches more homogeneous conditions.

3.2 Reynolds Stresses

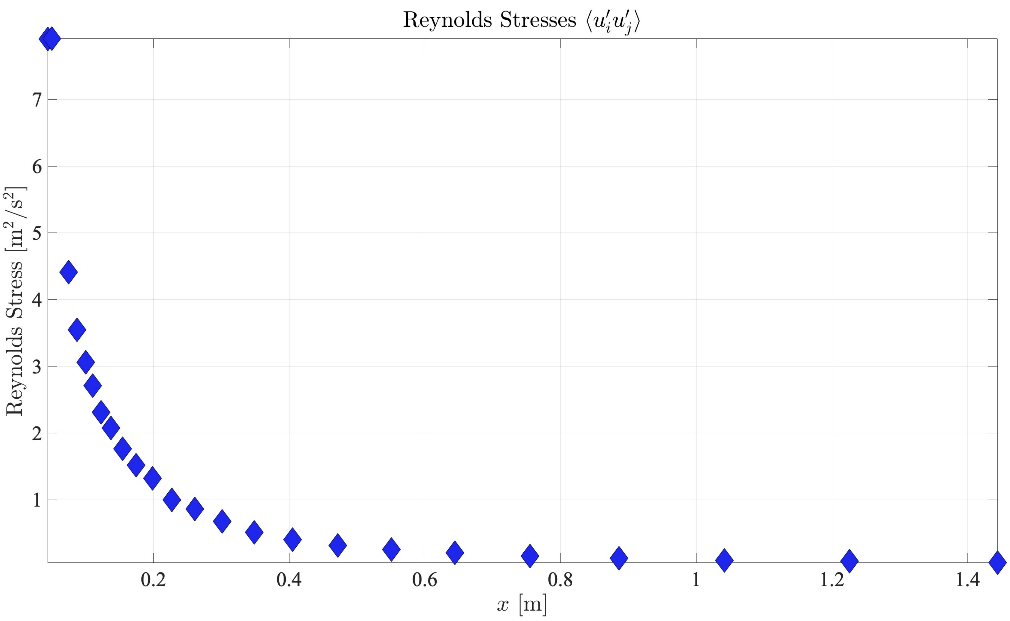

FIGURE 7 – Reynolds Stresses along the centerline

The plot of Reynolds stresses $\langle u'_i u'_j \rangle$ shows a rapid decay of turbulence intensity along the centerline, with high values near the grid that diminish as x increases, approaching nearly zero by $x = 1.4\,\mathrm{m}$. This indicates that the flow is not homogeneous near the grid but tends to become more uniform downstream. However, without data on other velocity components (e.g., $\langle v'^2 \rangle$, $\langle u'v' \rangle$), it is difficult to assess isotropy conclusively. The rapid decrease in turbulence intensity suggests that isotropy and homogeneity might improve further downstream, but the flow remains anisotropic and inhomogeneous near the grid.

3.3 Taylor's Hypothesis

In turbulent flows, we often deal with different time scales for various phenomena. The characteristic time scales are defined as follows :

— **Advection Time Scale** : This is the time taken for the flow to advect a distance M (a characteristic length scale). It can be expressed as :

$$t_{\mathrm{adv}} = \frac{M}{U_\infty}$$

where U_∞ is the mean flow velocity.

— **Viscous Time Scale** : This is the time scale associated with viscous effects, given by :

$$t_{\text{vis}} = \frac{M^2}{\nu}$$

where ν is the kinematic viscosity of the fluid.

— **Turbulent Time Scale** : This time scale is related to the fluctuations in the turbulent flow, represented by :

$$t_{\text{turb}} = \frac{M}{u'}$$

where u' is the root mean square of the turbulent velocity fluctuations.

we can observe that the turbulent time scale t_{turb} is significantly shorter than the viscous time scale t_{vis}. Therefore, under certain conditions, we can make the following assumption :

$$t_{\text{turb}} \ll t_{\text{vis}} \text{ and } t_{\text{adv}}$$

This assumption allows us to focus primarily on the effects of turbulent advection when analyzing turbulent flows.

So, We decompose the velocity field into its mean and fluctuating components :

$$U(x,t) = U_\infty + u'(x,t)$$

where :

— U_∞ is the mean flow velocity.

— $u'(x,t)$ represents the fluctuating part of the velocity.

After that, We consider the momentum conservation equation (or the simplified Navier-Stokes equation for incompressible flow) :

$$\frac{\partial U}{\partial t} + U\frac{\partial U}{\partial x} = 0$$

Substituting the velocity decomposition into the momentum equation yields :

$$\frac{\partial}{\partial t}(U_\infty + u') + (U_\infty + u')\frac{\partial}{\partial x}(U_\infty + u') = 0$$

Now we expand each term :

$$\frac{\partial}{\partial t}(U_\infty + u') = \frac{\partial U_\infty}{\partial t} + \frac{\partial u'}{\partial t}$$

Since U_∞ is constant with respect to time, we have :

$$\frac{\partial U_\infty}{\partial t} = 0 \quad \Rightarrow \quad \frac{\partial}{\partial t}(U_\infty + u') = \frac{\partial u'}{\partial t}$$

$$(U_\infty + u')\frac{\partial}{\partial x}(U_\infty + u') = (U_\infty + u')\left(\frac{\partial U_\infty}{\partial x} + \frac{\partial u'}{\partial x}\right)$$

Again, since U_∞ is uniform, we find :

$$(U_\infty + u')\frac{\partial}{\partial x}(U_\infty + u') = (U_\infty + u')\frac{\partial u'}{\partial x}$$

The momentum equation thus simplifies to :

$$\frac{\partial u'}{\partial t} + (U_\infty + u')\frac{\partial u'}{\partial x} = 0$$

If we assume that the fluctuations are small compared to the mean flow, such that $u' \ll U_\infty$, we can neglect the term $u'\frac{\partial u'}{\partial x}$ in the equation. Thus, we obtain :

$$\frac{\partial u'}{\partial t} + U_\infty \frac{\partial u'}{\partial x} = 0$$

Using Taylor's hypothesis, which assumes that the turbulent fluctuations are frozen in space as they move downstream, we replace the temporal derivative with a spatial derivative. We relate the time and space derivatives as follows :

$$\frac{\partial u'}{\partial t} = \frac{du'}{dx}\frac{dx}{dt} = U_\infty \frac{\partial u'}{\partial x}$$

This leads to :

$$\frac{du'}{dx} + U_\infty \frac{\partial u'}{\partial x} = 0$$

3.4 TKE Transport Equation

Assuming transport terms by pressure and turbulence diffusion are negligible, show that the transport equation of the turbulent kinetic energy (TKE) $k = \langle u_i' u_i' \rangle /2$ along the centerline becomes

$$-U\frac{dk}{dx} - (\langle u'^2 \rangle - \langle v'^2 \rangle)\frac{dU}{dx} - \epsilon = 0,$$

where ϵ is the mean dissipation rate of the turbulent kinetic energy. Which of these quantities can be calculated using the experimental set-up ? What does the intermediate term represent ? What about its streamwise evolution ?

The general transport equation for turbulent kinetic energy (TKE) is :

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = \mathcal{P} - \varepsilon$$

where $k = \frac{1}{2}\langle u_i' u_i' \rangle$ is the turbulent kinetic energy, $\mathcal{P}$ represents the production term, and ε is the dissipation.

The production term $\mathcal{P}$ is given by :

$$\mathcal{P} = -R_{ij}\frac{\partial U_i}{\partial x_j}$$

where $R_{ij} = \langle u_i' u_j' \rangle$ is the Reynolds stress tensor, which in 2D is written as :

$$R_{ij} = \begin{bmatrix} \langle u'^2 \rangle & \langle u'v' \rangle \\ \langle u'v' \rangle & \langle v'^2 \rangle \end{bmatrix}$$

The mean velocity gradient in 2D is :

$$\frac{\partial U_i}{\partial x_j} = \begin{bmatrix} \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} \end{bmatrix}$$

Under the assumption $\langle u'v' \rangle \approx 0$, the production term simplifies to :

$$\mathcal{P} = \langle u'^2 \rangle \frac{\partial U}{\partial x} + \langle v'^2 \rangle \frac{\partial V}{\partial y}$$

In the stationary case ($\frac{\partial k}{\partial t} = 0$) and neglecting the turbulent diffusion and pressure terms, the transport equation becomes :

$$U\frac{dk}{dx} = \mathcal{P} - \varepsilon$$

Substituting $\mathcal{P} = \langle u'^2 \rangle \frac{\partial U}{\partial x} - \langle v'^2 \rangle \frac{\partial U}{\partial x}$, we get :

$$U\frac{dk}{dx} = \langle u'^2\rangle\frac{\partial U}{\partial x} - \langle v'^2\rangle\frac{\partial U}{\partial x} - \varepsilon$$

Using the assumption $\frac{dU}{dx} = -\frac{dV}{dy}$, leading to the final equation :

$$U\frac{dk}{dx} = \langle u'^2\rangle\frac{\partial U}{\partial x} + \langle v'^2\rangle\frac{\partial U}{\partial x} - \varepsilon$$

Thus,

$$-U\frac{dk}{dx} - (\langle u'^2\rangle\frac{\partial U}{\partial x} - \langle v'^2\rangle\frac{\partial U}{\partial x}) - \varepsilon = 0$$

This equation highlights the balance between the advection of turbulent kinetic energy, its production in both the x and y directions, and the dissipation ε.

3.5 TKE Decay Law

In HIT, the TKE is expected to decay as

$$\frac{k}{U^2} = A\left(\frac{x-x_0}{M}\right)^{-m},$$

with x_0 being the so-called virtual origin. Identify the region where this law is valid and calculate x_0, the constant A, and the power law exponent m.

3.6 Dissipation Rate and Kolmogorov Length Scale

Estimate ϵ in the nearly HIT region and compare it to its isotropic formulation $\epsilon_{\text{iso}} = 15\nu\langle(\partial u'/\partial x)^2\rangle$. Deduce the Kolmogorov length scale η. Is the flow locally isotropic ?

4 Two-Point Statistics

4.1 Two-Point Correlation Tensor

In HIT, the two-point correlation tensor $R_{ij}(\vec{x}, r, t) = \langle u'_i(\vec{x}, t)u'_j(\vec{x} + r, t)\rangle$ can be expressed as

$$R_{ij}(r,t) = \langle u'^2\rangle\left[g(r)\delta_{ij} + (f(r) - g(r))\,\frac{r_i r_j}{r^2}\right],$$

where f and g are the longitudinal and transverse autocorrelation coefficients, respectively, and r_j is the j-th component of the spatial separation r, such that $r^2 = r_j r_j$. Show that the continuity equation induces $\partial R_{ij}/\partial r_i = \partial R_{ij}/\partial r_j = 0$.

4.2 Longitudinal and Transverse Correlation

Deduce that $g(r) = f(r) + \frac{r}{2}\frac{df}{dr}$. Comment and check the validity of this expression on your data.

4.3 Integral Length Scale

Compute the integral length scale and compare it to η.

4.4 Energy Spectrum

Compute the energy spectrum and comment (plot cumulative energy and dissipative spectrum).